

HydroNet-Proof

Deterministic Mathematical Framework for Fluid Dynamics & Hydraulic Stress Allocation

1. Project Overview

This technical framework establishes a rigorous, manual mathematical model to evaluate steady-state fluid mechanics, volumetric mass balance, and energy dissipation profiles across closed-conduit infrastructure networks. By bypassing automated commercial simulation software, this work focuses entirely on step-by-step mathematical proofs, symbolic differentiation, and manual numerical analysis convergence techniques required to map localized hydraulic boundary stress and pressure distributions.

2. Governing Continuum Mechanics

The internal network routing is systematically evaluated as a 1D incompressible steady-state flow along a discrete spatial coordinate axis. The system strictly enforces fundamental physical conservation laws:

Conservation of Mass (Continuity Theory)

For an internal conduit network characterized by variable cross-sectional geometric boundaries, the volumetric mass flow rate must remain invariant across all localized intersection nodes:

$$A_1 \cdot v_1 = A_2 \cdot v_2 = Q$$

Where the localized boundary cross-sectional area (A) is derived explicitly from the internal hydraulic pipe diameter (D):

$$A = (\pi \cdot D^2) / 4$$

Conservation of Momentum (Frictional Head Loss)

Energy degradation, pressure gradients, and frictional head loss (h_f) across the continuous spatial network coordinate system are analytically solved using the Darcy-Weisbach framework:

$$h_f = f \cdot (L / D) \cdot (v^2 / 2g)$$

Where f represents the dimensionless Darcy friction factor, L is the absolute linear length of the pipe segment (m), D is the hydraulic diameter (m), v is the mean fluid flow velocity (m/s), and g is the standard acceleration due to gravity (9.81 m/s²).

3. Boundary Conditions & Numerical Estimation Proofs

To establish the exact Darcy friction factor (f), multi-phase flow regimes must be dynamically classified via the dimensionless Reynolds Number (Re), calculating the fluid's inertial forces relative to its internal viscous shear stress bounds:

$$Re = (\rho \cdot v \cdot D) / \mu$$

Where ρ is fluid density (Water = 1000 kg/m^3) and μ is the absolute dynamic viscosity threshold ($1.002 \times 10^{-3} \text{ Pa}\cdot\text{s}$ at 20°C). Under highly turbulent boundary layers ($Re > 4000$), the friction factor becomes entirely implicit and transcendental, governed strictly by the Colebrook-White function. This framework maps the manual optimization derivation via the Newton-Raphson iterative root-finding methodology:

Implicit Objective Function Formulation

$$F(f) = 1 / \sqrt{f} + 2 \cdot \log_{10} [(\epsilon / (3.71 \cdot D)) + (2.51 / (Re \cdot \sqrt{f}))] = 0$$

Where ϵ denotes the absolute structural roughness parameter of the physical conduit boundary layer.

First-Order Symbolic Differentiation Proof

To systematically execute successive linear approximations using the classical iterative step $f_{n+1} = f_n - [F(f_n) / F'(f_n)]$, the exact analytical first derivative is manually established through standard calculus chain rules:

$$F'(f) = -0.5 \cdot f^{-1.5} - [2 / \ln(10)] \cdot [(\epsilon / (3.71 \cdot D)) + (2.51 / (Re \cdot \sqrt{f}))]^{-1} \cdot [-1.255 / (Re \cdot f^{-1.5})]$$

4. Methodology & Analytical Application Scope

- **Analytical Derivations:** Executed manual algebraic expansion of multi-variable system continuity parameters across high-density network intersections.
- **Parametric Sensitivity Analysis:** Formulated deterministic proofs illustrating how minor variances in boundary wall roughness (ϵ) scale non-linearly against downstream volumetric flow velocities and continuous mechanical head drops.
- **Structural Boundary Estimation:** Provided quantitative mappings of localized pressure depletion spikes to isolate hydraulic pipe fracturing risks and infrastructure fatigue thresholds.